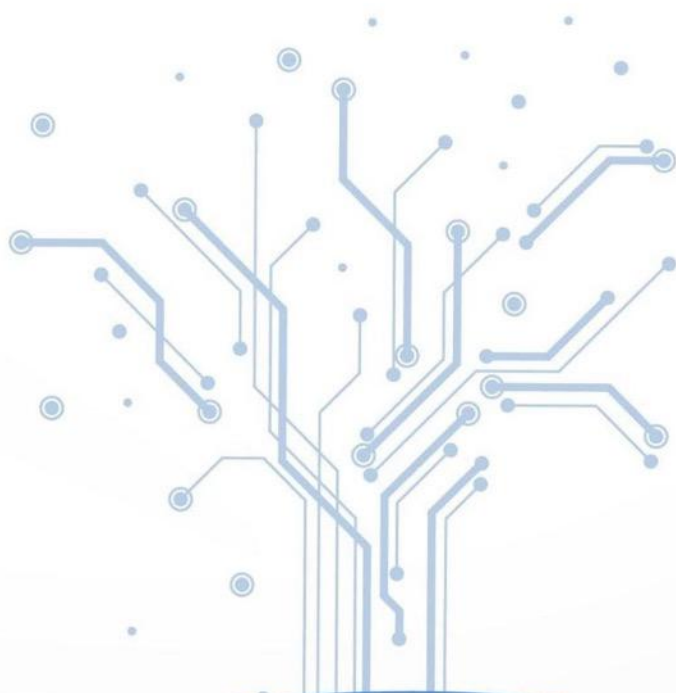




Animal visit reporting Software & Mobile app specifications



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This project has received funding from the European Union's Horizon Europe research and innovation programme under grant agreement No 101134083

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Title: Democratising digital farming through tailored open source and open hardware solutions

Type of action: HORIZON Research and Innovation Action

Topic: HORIZON-CL6-2023-GOVERNANCE-01-13

Start date of project: 1 January 2024 • Duration: 36 months

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1 Software & Mobile app specifications

1.1 OpenAgri online software (Cloud mode)

Software technologies

Web:	• PostgreSQL database • PHP framework Laravel
Mobile App:	• Flutter • ISAR database
UHF case:	• JDK for embedded development • SQL lite

1.2 OpenAgri mobile App (Edge mode)

Software Title: OpenAgri mobile app

Short Description: Mobile application to collect UHF Data from Page Up UHF cases and to compute locally visits and build daily visits reports.

Keywords/Tags: RFID UHF, livestock electronic identification, visits to areas of interest

Author(s)/Maintainer(s): F. Gimbert

Organization: Page Up

Contact Email: fgimbert@pageup.fr

Repository URL: (add registry url for docker images if exist)

Version: 1.0.0

Programming Language(s): Flutter

Operating System(s): Android 10+

System architecture: The mobile application is a standalone app (no backoffice nor web site needed). To collect data, it must connect, via bluetooth, to a PageUp UHF box.

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2 REST API specification

The following specification refers to the webservice of the RFID Data Logger.

/api/rows/page		
This endpoint allows querying read tags given several filtering/selection options. Data is paginated.		
Request		
start_datetime	Tags read after given date	"2025-03-03 03:03:03"
end_datetime	Tags read before given date	"2025-03-03 06:06:06"
case_ids	Array of readers ids	{ "CASE-A226", "CASE-A221" }
antenna_ids	Array of antenna ids. Based on the nomenclature of the Impinj reader.	{ "1", "2", "3" }
site_ids	Array of site ids	{ "Magny", "Jalogny" }
tags	Array of tags	{ "8200E92838B7" }
Response		
<pre>{ "filters": { "tag_id": null, "case_ids": [], "site_ids": ["Le Rheu"], "antenna_ids": [], "start_datetime": "2024-07-01 00:00:00", "end_datetime": "2025-03-12 00:00:00", "groups": [], "tags": [] }, "parameters": { "page": 1, "per_page": 10, "sort": "created_at", "order": "desc" }, "pagination": { "total": 88374, "per_page": 10, "current_page": 1, "last_page": 8838, "from": 1, </pre>		

```
"to": 10
},
"items": [
  {
    "id": 37917614,
    "tag_id": "8200F800EE6B294E8603004E01000000",
    "reader_id": "AE3F22-A226",
    "antenna_id": "4",
    "read_count": 34,
    "rssi": "-51",
    "created_at": "2024-10-23 14:50:58",
    "updated_at": null,
    "deleted_at": null,
    "rssi_min": "-63.5",
    "rssi_max": "-51",
    "rssi_average": "-53.941177",
    "site_id": 3,
    "computed_id": "992 4000000334",
    "software_version": "1.10-SNAPSHOT",
    "utc_date_time": "2024-10-23 14:51:05",
    "stored_at": "2024-10-23 16:51:02",
    "readingdate": "2024-10-23 16:51:05",
    "not_utc": false
  },
  {
    "id": 37917613,
    "tag_id": "8200F800EE6B294E8603004E01000000",
    "reader_id": "AE3F22-A226",
    "antenna_id": "4",
    "read_count": 34,
    "rssi": "-48.5",
    "created_at": "2024-10-23 14:50:57",
    "updated_at": null,
    "deleted_at": null,
    "rssi_min": "-63.5",
    "rssi_max": "-51",
    "rssi_average": "-53.941177",
    "site_id": 3,
    "computed_id": "992 4000000334",
    "software_version": "1.10-SNAPSHOT",
    "utc_date_time": "2024-10-23 14:51:06",
    "stored_at": "2024-10-23 16:51:02",
    "readingdate": "2024-10-23 16:51:06",
    "not_utc": false
  }
]
}
```

/api/cases/page		
This endpoint allows querying available RFID cases declared in the application.		
Request		
name	Exact or partial name of RFID cases.	"A226"
Response		
<pre>{ "filters": { "name": null, "status": null }, "parameters": { "page": 1, "per_page": 10, "sort": "id", "order": "asc" }, "pagination": { "total": 23, "per_page": 10, "current_page": 1, "last_page": 3, "from": 1, "to": 10 }, "items": [{ "id": 8, "name": "CASE-A220", "code": "AE3F22-A220", "lastcomm_datetime": "2023-08-06 08:18:43+00", "description": "Cloud, projet TechCare", "status": false, "site_id": 10, "created_at": "2022-02-07 06:55:43", "updated_at": "2023-08-06 09:20:01", "deleted_at": null, "monitor": true, "antennas_count": null, "monitoring_tags": 1, "site_name": "NIBIO", "favorite": 1, "softwarestatus": 0, "sites": null }] }</pre>		

```
}  
]  
}
```


/api/sites/page		
This endpoint allows querying available sites declared in the application.		
Request		
name	Exact or partial name of sites	"JALO"
country	Exact or partial name of country the site is declared in.	"FR"
Response		
<pre>{ "filters": { "name": null, "country": null, "address": null, "include": [] }, "parameters": { "page": 1, "per_page": 20, "sort": "id", "order": "asc" }, "pagination": { "total": 12, "per_page": 20, "current_page": 1, "last_page": 1, "from": 1, "to": 12 }, "items": [{ "id": 1, "name": "Page Up", "country": "France", "address": "13 rue de Dijon", "description": "description 1", "created_at": null, "updated_at": null, "deleted_at": null }, { "id": 2, "name": "Site 2", "country": "France",</pre>		

```
"address": "10 rue de Dijon",  
"description": "description 2",  
"created_at": null,  
"updated_at": null,  
"deleted_at": null  
},
```

/api/visits/page		
This endpoint allows querying visits computed by the application.		
Request		
start_datetime	Tags read after given date	"2025-03-03 03:03:03"
end_datetime	Tags read before given date	"2025-03-03 06:06:06"
case_ids	Array of readers ids	{ "CASE-A226", "CASE-A221" }
antenna_ids	Array of antenna ids. Based on the nomenclature of the Impinj reader.	{ "1", "2", "3" }
site_ids	Array of site ids	{ "Magny", "Jalogny" }
tags	Array of tags	{ "8200E92838B7" }
Response		
<pre>{ "filters": { "name": null, "country": null, "address": null, "include": [] }, "parameters": { "page": 1, "per_page": 20, "sort": "id", "order": "asc" }, "pagination": { "total": 12, "per_page": 20, "current_page": 1, "last_page": 1, "from": 1, "to": 12 }, "items": [{ ... }] }</pre>		

```
}  
]  
}
```